# Supplementary Information

Independent Real-Time Verification of Manual Valve Position as an Additional Protection Layer Against Misrouting During Product Transfers: A Monte Carlo Barrier-Effectiveness Assessment

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## S1. First suite: all outputs for the 14-valve route

| Output | Existing checks | Turn-count collar | Direct sensing | Direct + NFC |
|---|---|---|---|---|
| **Undetected error per transfer** | 4.027% (3.957–4.098%) | 0.454% (0.431–0.479%) | 0.302% (0.283–0.322%) | 0.285% (0.267–0.305%) |
| **from a valve position** | 3.206% | 0.414% | 0.264% | 0.247% |
| **from an out-of-sequence step** | 0.346% | 0.026% | 0.026% | 0.026% |
| **from a post-validation movement** | 0.503% | 0.015% | 0.013% | 0.013% |
| **Validation with at least one nuisance alarm** | 0.0% | 54.1% | 18.6% | 17.5% |
| **Validation blocked (device unavailable)** | 0.0% | 8.0% | 8.0% | 8.0% |
| **Bypass of a blocked validation** | 0.00% | 1.61% | 1.61% | 1.61% |
| **Nuisance alarms per valve** | 0.00% | 5.37% | 1.42% | 1.33% |
| **Defects missed by the sensors** | – | 8.5% | 5.5% | 5.6% |
| **Detection of a post-validation movement within 60 s** | – | 96.6% | 97.0% | 97.0% |
| **Wrong-valve events per 1000 transfers** | 20.6 | 20.6 | 20.6 | 3.0 |

*Table S1. Outputs of the first suite for the 14-valve route (300,000 transfers per cell; Wilson 95% interval for the primary output).*

| Tolerance | Direct: alarms | Direct: missed | Turns: alarms | Turns: missed | Direct aged: alarms | Direct aged: missed | Turns aged: alarms | Turns aged: missed |
|---|---|---|---|---|---|---|---|---|
| **3%** | 26.7% | 1.7% | 43.4% | 2.1% | 28.3% | 1.9% | 61.7% | 4.0% |
| **5%** | 7.9% | 2.0% | 20.4% | 3.4% | 9.6% | 2.3% | 43.1% | 6.3% |
| **6%** | 4.3% | 2.4% | 13.2% | 4.7% | 5.9% | 2.8% | 36.0% | 7.9% |
| **8%** | 1.4% | 5.6% | 5.4% | 8.3% | 2.9% | 5.8% | 25.9% | 11.8% |
| **10%** | 0.5% | 12.2% | 2.4% | 14.1% | 1.9% | 13.0% | 19.9% | 16.6% |
| **12%** | 0.3% | 22.2% | 1.4% | 22.0% | 1.5% | 22.4% | 16.3% | 22.5% |
| **15%** | 0.2% | 35.9% | 0.9% | 33.7% | 1.2% | 35.5% | 12.8% | 31.2% |

*Table S2. Alarm tolerance sweep (nuisance alarms per valve, share of true defects missed); aged = 60 operations since calibration.*

| Operations | Direct: alarms per valve | Direct: missed | Turns: alarms per valve | Turns: missed |
|---|---|---|---|---|
| **1** | 1.4% | 5.4% | 4.5% | 8.5% |
| **3** | 1.4% | 5.6% | 5.4% | 8.3% |
| **10** | 1.6% | 5.6% | 8.2% | 9.3% |
| **30** | 2.1% | 5.9% | 15.8% | 10.7% |
| **100** | 3.9% | 6.0% | 36.9% | 13.6% |
| **300** | 8.7% | 7.0% | 64.7% | 16.2% |

*Table S3. Operations since calibration (tolerance 8%).*

| Communication loss | Validation blocked | Residual risk, 20% bypass | Residual risk, no bypass |
|---|---|---|---|
| **0.0005** | 4.8% | 0.245% | 0.208% |
| **0.0010** | 5.4% | 0.250% | 0.208% |
| **0.0030** | 8.0% | 0.274% | 0.208% |
| **0.0100** | 16.8% | 0.343% | 0.208% |
| **0.0200** | 27.8% | 0.454% | 0.208% |
| **0.0500** | 53.4% | 0.709% | 0.208% |

*Table S4. Per-device communication-loss probability: validation blocked, and residual risk of the full layer with 20% and 0% bypass.*

| Complacency | Residual risk |
|---|---|
| **0%** | 0.229% |
| **10%** | 0.253% |
| **20%** | 0.274% |
| **40%** | 0.325% |
| **60%** | 0.378% |
| **80%** | 0.431% |

*Table S5. Operator complacency (relative reduction of the effectiveness of existing checks) and residual risk of the full layer.*

| Tap compliance | Wrong-valve events per 1000 |
|---|---|
| **0%** | 20.4 |
| **25%** | 15.8 |
| **50%** | 11.0 |
| **75%** | 6.2 |
| **90%** | 3.1 |
| **100%** | 1.1 |

*Table S6. NFC tap compliance and wrong-valve events per 1000 transfers.*

| Parameter | Half | Double |
|---|---|---|
| **existing-barrier miss rate** | 0.18% | 0.46% |
| **skip rate** | 0.27% | 0.28% |
| **incomplete-movement rate** | 0.25% | 0.32% |
| **out-of-sequence rate** | 0.26% | 0.29% |
| **unauthorized-movement rate** | 0.27% | 0.28% |
| **sensor noise** | 0.23% | 0.39% |
| **calibration bias** | 0.18% | 0.48% |
| **alarm tolerance** | 0.09% | 1.96% |
| **device unavailability** | 0.24% | 0.33% |
| **bypass probability** | 0.24% | 0.36% |
| **operator complacency** | 0.25% | 0.33% |
| **wrong-device assignment** | 0.27% | 0.27% |

*Table S7. One-at-a-time sensitivity of the residual risk of the full layer (base 0.27%) for the parameter at half and double its base value.*

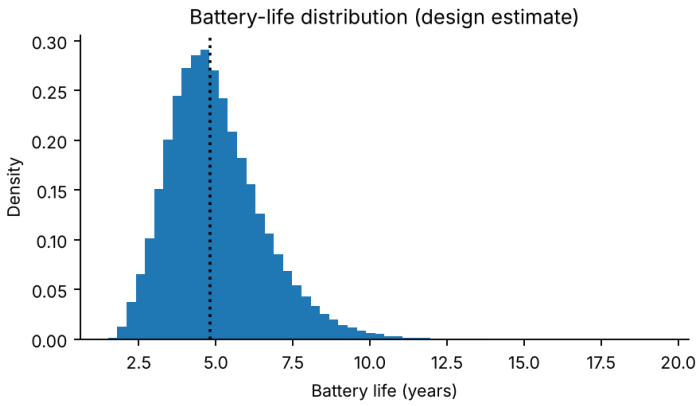



*Fig. S1. Design-level battery-life estimate (primary lithium cells, wake-on-motion, 16 s heartbeat): median 4.8 years, 5th to 95th percentile 2.9 to 7.9 years; the analytic mean is 4.7 years.*

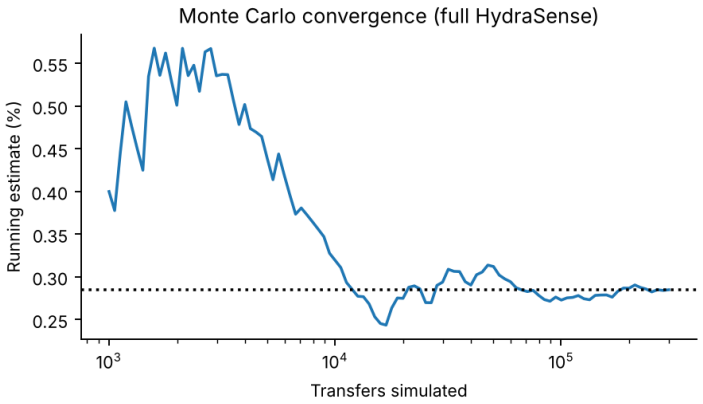



*Fig. S2. Convergence of the Monte Carlo estimate of the residual risk with the number of transfers.*

## S2. Extended studies: all cases

| Response | Fatigue coefficient | Residual risk | Risk-reduction factor |
|---|---|---|---|
| **100%** | 0.0 | 0.275% (0.249–0.302%) | 14.8 |
| **100%** | 0.3 | 0.454% (0.421–0.489%) | 8.9 |
| **100%** | 0.6 | 0.591% (0.553–0.631%) | 6.9 |
| **90%** | 0.0 | 0.720% (0.678–0.764%) | 5.6 |
| **90%** | 0.3 | 0.884% (0.838–0.933%) | 4.6 |
| **90%** | 0.6 | 1.009% (0.960–1.061%) | 4.0 |
| **80%** | 0.0 | 1.145% (1.092–1.200%) | 3.5 |
| **80%** | 0.3 | 1.301% (1.245–1.360%) | 3.1 |
| **80%** | 0.6 | 1.405% (1.347–1.466%) | 2.9 |

*Table S8. Operator response to an isolated alarm, fatigue coefficient, residual risk of the full layer (Wilson 95% interval) and risk-reduction factor (existing checks 4.06%).*

| Response | Fatigue | Tolerance | Residual risk | Nuisance alarms per validation |
|---|---|---|---|---|
| **100%** | 0.0 | 4% | 0.089% | 2.06 |
| **100%** | 0.0 | 5% | 0.102% | 1.10 |
| **100%** | 0.0 | 6% | 0.124% | 0.59 |
| **100%** | 0.0 | 7% | 0.169% | 0.33 |
| **100%** | 0.0 | 8% | 0.275% | 0.19 |
| **100%** | 0.0 | 10% | 0.625% | 0.06 |
| **100%** | 0.0 | 12% | 1.132% | 0.02 |
| **100%** | 0.0 | 15% | 1.791% | 0.01 |
| **95%** | 0.3 | 4% | 2.079% | 2.06 |
| **95%** | 0.3 | 5% | 1.355% | 1.10 |
| **95%** | 0.3 | 6% | 0.898% | 0.59 |
| **95%** | 0.3 | 7% | 0.707% | 0.33 |
| **95%** | 0.3 | 8% | 0.661% | 0.19 |
| **95%** | 0.3 | 10% | 0.878% | 0.06 |
| **95%** | 0.3 | 12% | 1.315% | 0.02 |
| **95%** | 0.3 | 15% | 1.936% | 0.01 |
| **95%** | 0.6 | 4% | 2.939% | 2.06 |
| **95%** | 0.6 | 5% | 1.945% | 1.10 |
| **95%** | 0.6 | 6% | 1.277% | 0.59 |
| **95%** | 0.6 | 7% | 0.929% | 0.33 |
| **95%** | 0.6 | 8% | 0.788% | 0.19 |
| **95%** | 0.6 | 10% | 0.927% | 0.06 |
| **95%** | 0.6 | 12% | 1.329% | 0.02 |

| Response | Fatigue | Tolerance | Residual risk | Nuisance alarms per validation |
|---|---|---|---|---|
| **95%** | 0.6 | 15% | 1.943% | 0.01 |

*Table S9. Alarm tolerance with operator response: residual risk of the full layer and nuisance alarms per validation.*

| Binary tol. | Intermediate tol. | Alarms per binary valve | Alarms per intermediate valve | Alarms per validation | Defects missed | Residual risk |
|---|---|---|---|---|---|---|
| **5%** | 8% | 4.67% | 8.10% | 0.722 | 5.3% | 0.295% |
| **5%** | 12% | 4.67% | 0.91% | 0.578 | 20.1% | 1.063% |
| **5%** | 15% | 4.67% | 0.33% | 0.567 | 30.0% | 1.578% |
| **6%** | 8% | 1.81% | 8.10% | 0.379 | 5.3% | 0.297% |
| **6%** | 12% | 1.81% | 0.91% | 0.235 | 20.1% | 1.064% |
| **6%** | 15% | 1.81% | 0.33% | 0.223 | 30.0% | 1.579% |
| **8%** | 8% | 0.31% | 8.10% | 0.199 | 5.6% | 0.312% |
| **8%** | 12% | 0.31% | 0.91% | 0.055 | 20.4% | 1.079% |
| **8%** | 15% | 0.31% | 0.33% | 0.044 | 30.3% | 1.594% |

*Table S10. Separate tolerances for binary and intermediate valves (direct sensing, 14-valve route).*

| Operations | Calibration | Nuisance alarms per valve | Defects missed |
|---|---|---|---|
| **1** | calibration at assignment | 1.37% | 5.2% |
| **1** | re-zero at end stops | 1.36% | 5.2% |
| **3** | calibration at assignment | 1.42% | 5.6% |
| **3** | re-zero at end stops | 1.37% | 5.6% |
| **10** | calibration at assignment | 1.60% | 5.7% |
| **10** | re-zero at end stops | 1.39% | 5.7% |
| **30** | calibration at assignment | 2.11% | 5.8% |
| **30** | re-zero at end stops | 1.47% | 5.7% |
| **100** | calibration at assignment | 3.88% | 5.8% |
| **100** | re-zero at end stops | 1.72% | 5.6% |
| **300** | calibration at assignment | 8.69% | 7.2% |
| **300** | re-zero at end stops | 2.35% | 6.5% |

*Table S11. End-stop re-zeroing (direct sensing).*

| Coverage | Undetected, full layer | Undetected, existing checks | Events, with NFC | Events, without NFC |
|---|---|---|---|---|
| **0%** | 0.83 | 5.07 | 1.38 | 10.11 |
| **25%** | 0.65 | 5.07 | 1.38 | 10.11 |
| **50%** | 0.45 | 5.07 | 1.38 | 10.11 |
| **75%** | 0.23 | 5.07 | 1.38 | 10.11 |
| **100%** | 0.01 | 5.07 | 1.38 | 10.11 |

*Table S12. Coverage of neighboring valves: undetected collateral events per 1000 transfers (full layer and existing checks), and collateral events (with and without NFC tap-before-touch).*

| Log-standard deviation | Existing checks (95% interval) | Full layer (95% interval) | Risk-reduction factor |
|---|---|---|---|
| **0.0** | 4.06% (3.99–4.13%) | 0.276% (0.258–0.295%) | 14.7 |
| **0.5** | 4.04% (3.97–4.11%) | 0.276% (0.258–0.295%) | 14.7 |
| **1.0** | 4.02% (3.95–4.10%) | 0.275% (0.257–0.294%) | 14.6 |
| **1.5** | 3.92% (3.85–3.99%) | 0.273% (0.255–0.293%) | 14.3 |

*Table S13. Common cause: log-standard deviation of the shared multiplier of the operator error rates.*

| Input | Risk-reduction factor | Full layer risk | Existing-checks risk |
|---|---|---|---|
| **existing-barrier miss rate (x)** | 0.64 | 0.73 | 0.96 |
| **sensor calibration bias (x)** | −0.61 | 0.61 | −0.03 |
| **device unavailability (x)** | −0.55 | 0.54 | 0.05 |
| **sensor noise (x)** | −0.50 | 0.54 | −0.04 |
| **bypass probability** | −0.45 | 0.52 | 0.05 |

| Input | Risk-reduction factor | Full layer risk | Existing-checks risk |
|---|---|---|---|
| operator error rates (x) | 0.42 | 0.59 | 0.91 |
| operator complacency | −0.38 | 0.44 | 0.03 |
| time-stamp sequence detection | 0.28 | −0.32 | 0.06 |
| unauthorized-movement rate (x) | 0.25 | 0.06 | 0.46 |
| sensor drift (x) | −0.10 | 0.06 | −0.02 |
| out-of-sequence rate (x) | −0.09 | 0.27 | 0.32 |
| wrong-device assignment (x) | −0.07 | −0.04 | −0.08 |
| slip rate (x) | −0.02 | −0.04 | −0.01 |

*Table S14. Partial rank correlation coefficients of the 13 assumed inputs with the logarithm of the risk-reduction factor, of the residual risk of the full layer and of the residual risk of existing checks.*

| Readings | Lower | Upper |
|---|---|---|
| 30 | 0.796 | 1.344 |
| 50 | 0.835 | 1.246 |
| 100 | 0.878 | 1.162 |
| 200 | 0.911 | 1.109 |
| 500 | 0.942 | 1.066 |

*Table S15. 95% interval for the standard deviation of repeated readings, as a multiple of the observed value.*

## S3. Reproducibility

The repository (https://github.com/sandlerleon/hydrasense-valve-verification-mc, version 1.0.0) contains the unchanged first-suite model (HydraSense_MC.py), the extended engine and studies (hydrasense_v2.py), the results (results/original_results.json, results/results_v2.json), the figures, the 12 automated tests (tests/test_all.py) and the manuscript sources. Commands: python HydraSense_MC.py --reproduce reproduces the first suite; python hydrasense_v2.py --all runs the extended studies (about 4.5 min on one core), python tests/test_all.py runs the tests, python make_schematic.py draws Fig. 1. The seed of the extended studies is 2026; every study uses fixed seeds, so rerunning reproduces every number in the text. The code needs Python 3, numpy, scipy and matplotlib.